A SEMICLASSICAL BAND-AREA CORRESPONDENCE FOR THE AM–GM CONE

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Abstract. Working from the cone $X^2 + Y^2 = T^2$ developed in [1] and the exact $\mathfrak{su}(2)$ Schwinger-boson representation attached to it in [2], we compute in closed form the true (induced) surface area of the band of the cone lying between two consecutive anti-diagonals $x + y = K$ and $x + y = K + 2$, and compare its large-K growth against the semiclassical limit of the loop quantum gravity area-operator spectrum $A(j) \approx 8\pi\gamma\ell_P^2 j$. Identifying one lattice unit with one Planck length, the two grow with the same functional form in $j = (K - 1)/2$, and matching coefficients fixes an effective Barbero–Immirzi parameter $\gamma_{\text{eff}} = 1/(4\sqrt{2}) \approx 0.177$. We record this as a self-contained, fully worked consistency check, and are explicit about what it does and does not establish: it is a coincidence of normalization, not a derivation of γ, and it does not repair the unresolved mismatch between a single edge of the construction (an entire anti-diagonal) and a single quantum of area (a puncture).

1. Setup

We use the coordinates and conventions of [1]: for a factor pair (x, y) of positive reals,

$$X = \frac{x - y}{2}, \qquad Y = \sqrt{xy}, \qquad T = \frac{x + y}{2},$$

satisfying $X^2+Y^2 = T^2$. Fixing the anti-diagonal $x+y = K$ fixes $T = K/2$; under the oscillator dictionary of [2, §2], $K = 2j + 1$ identifies this anti-diagonal with the spin-j irreducible representation realized by the Schwinger bosons of [2, §3], whose Casimir eigenvalue is $j(j + 1) = (T - \frac{1}{2})(T + \frac{1}{2})$.

Loop quantum gravity assigns to a single spin-network edge carrying label j a quantized area

$$A(j) = 8\pi\gamma\ell_P^2\sqrt{j(j + 1)}, \tag{1}$$

γ the Barbero–Immirzi parameter, ℓ_P the Planck length [3]. The present note asks a narrow, purely formal question: does the classical (differential-geometric) area of the region of the cone naturally associated to one anti-diagonal grow, in K, at the same rate as (1) grows in j – and if a matching normalization is imposed, what γ does it force?

2. The band area, exactly

Proposition 1. *The map $(x, y) \mapsto (X, Y)$ restricts, on the open strip $\{(x, y) : x, y > 0,\ K \le x+y \le K+2\}$, to a diffeomorphism onto the half-annulus $\{(X, Y) : T_1 \le \sqrt{X^2 + Y^2} \le T_2,\ Y \ge 0\}$, $T_1 = K/2$, $T_2 = (K + 2)/2$.*

Proof. For fixed $s = x + y$, the point $(X, Y) = \left(\frac{x-y}{2}, \sqrt{xy}\right)$ satisfies $X^2 + Y^2 = (s/2)^2$ identically, and as x ranges over $(0, s)$, X ranges monotonically over $(-s/2, s/2)$ with $Y = \sqrt{(s/2)^2 - X^2} \ge 0$: the full upper semicircle of radius $s/2$. Varying $s \in [K, K + 2]$ sweeps every such semicircle exactly once, giving the stated bijection; smoothness and non-vanishing of the Jacobian on $x, y > 0$ give the diffeomorphism. □

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By Proposition 1 and the change-of-variables theorem, the flat-shadow area of the strip is exactly the area of the half-annulus, $\frac{\pi}{2}(T_2^2 - T_1^2) = \frac{\pi}{2}(K+1)$, with no approximation and no lattice-cell summation required. The physically relevant quantity, however, is the true induced area on the cone as embedded in Euclidean 3-space, not its flat shadow.

Proposition 2. *Parametrizing the cone by $(T, \theta) \mapsto (T\sin\theta, T\cos\theta, T)$, $\theta \in [-\pi/2, \pi/2]$, the induced area element is $dA = \sqrt{2}\,T\,dT\,d\theta$, and the exact 3-dimensional surface area of the band between anti-diagonals K and $K+2$ is*

$$A_{\text{band}}(K) = \frac{\pi}{\sqrt{2}}\left(T_2^2 - T_1^2\right) = \frac{\pi}{\sqrt{2}}(K+1). \tag{2}$$

Proof. With $r(T,\theta) = (T\sin\theta, T\cos\theta, T)$, $r_T = (\sin\theta, \cos\theta, 1)$, $r_\theta = (T\cos\theta, -T\sin\theta, 0)$, and $|r_T \times r_\theta| = |(T\sin\theta,\, T\cos\theta,\, -T)| = T\sqrt{2}$. Integrating over $\theta \in [-\pi/2, \pi/2]$ (length π) and $T \in [T_1, T_2]$ gives (2); this agrees with the elementary frustum formula $\frac{1}{2}\pi(T_1+T_2)\cdot\sqrt{2}(T_2 - T_1)$ for a $45°$ half-angle cone restricted to its $Y \geq 0$ half. □

Equation (2) is exact for every $K \geq 1$, not merely asymptotic.

Remark 3 (An independent Cartesian cross-check)**.** *Proposition 2 computes dA via the cross product $|r_T \times r_\theta|$ in the polar parametrization (T,θ). As a check obtained by an unrelated route, pull back the embedding $(x,y) \mapsto (X,Y,T) = \left(\frac{x-y}{2}, \sqrt{xy}, \frac{x+y}{2}\right)$ directly in the original Cartesian factor-pair coordinates. Writing $E = X_x^2 + Y_x^2 + T_x^2$, $G = X_y^2 + Y_y^2 + T_y^2$, $F = X_xX_y + Y_xY_y + T_xT_y$ for the first fundamental form, direct computation gives*

$$E = \frac{2x+y}{4x}, \qquad G = \frac{x+2y}{4y}, \qquad F = \frac{1}{4}, \qquad EG - F^2 = \frac{(x+y)^2}{8xy},$$

so that

$$dA = \sqrt{EG - F^2}\,dx\,dy = \frac{x+y}{2\sqrt{2}\,\sqrt{xy}}\,dx\,dy = \frac{T}{\sqrt{2}\,Y}\,dx\,dy. \tag{2$'$}$$

This is the same area element as Proposition 2, in different coordinates: converting $(2')$ to (T,θ) via the Jacobian $\left|\partial(T,\theta)/\partial(x,y)\right| = 1/(2\sqrt{xy})$ (itself a direct computation from $T = \frac{x+y}{2}$ and $\theta = \arctan(X/Y)$) gives $dA = \sqrt{2}\,T\,dT\,d\theta$ exactly, matching Proposition 2 with no free constant to adjust. The local distortion factor $T/(\sqrt{2}\,Y)$ in $(2')$ is minimized, at value $1/\sqrt{2}$, exactly on the diagonal $x = y$, and grows without bound as a cell moves off the diagonal ($Y \to 0$ at fixed T) — so while $Y^2 = xy$ recovers a cell's flat area as a single number, the patch of cone surface that cell actually occupies is not uniform across the table, a fact this note's band-area computation already accounts for correctly since Proposition 2 integrates the exact element rather than assuming a constant one.

Proposition 4 (Exact area of a single lattice cell)**.** *For $x_0, y_0 \geq 0$, write $\Delta_{3/2}(t_0) := (t_0+1)^{3/2} - t_0^{3/2}$ and $\Delta_{1/2}(t_0) := \sqrt{t_0+1} - \sqrt{t_0}$. The true induced area on the cone of the image, under $(x,y) \mapsto (X,Y,T)$, of the unit cell $[x_0, x_0+1] \times [y_0, y_0+1]$ is*

$$A_{\text{cell}}(x_0,y_0) = \frac{\sqrt{2}}{3}\Big[\Delta_{3/2}(x_0)\,\Delta_{1/2}(y_0) + \Delta_{1/2}(x_0)\,\Delta_{3/2}(y_0)\Big], \tag{4}$$

exact in closed form, not merely asymptotic, and not requiring the local density $(2')$ to be constant across the cell.

Proof. By $(2')$, $dA = \frac{1}{2\sqrt{2}}\left(\sqrt{x/y} + \sqrt{y/x}\right)dx\,dy$. Fix x and integrate over $y \in [y_0, y_0+1]$: $\int \sqrt{x/y}\,dy = \sqrt{x}\cdot 2\sqrt{y}$ contributes $2\sqrt{x}\,\Delta_{1/2}(y_0)$, and $\int \sqrt{y/x}\,dy = \frac{1}{\sqrt{x}}\cdot\frac{2}{3}y^{3/2}$ contributes $\frac{2}{3\sqrt{x}}\Delta_{3/2}(y_0)$. Integrating the sum over $x \in [x_0, x_0+1]$: $\int 2\sqrt{x}\,dx = \frac{4}{3}x^{3/2}$ contributes $\frac{4}{3}\Delta_{3/2}(x_0)\,\Delta_{1/2}(y_0)$, and $\int \frac{2}{3\sqrt{x}}\,dx = \frac{4}{3}\sqrt{x}$ contributes $\frac{4}{3}\Delta_{1/2}(x_0)\,\Delta_{3/2}(y_0)$. Multiplying the sum by the prefactor $\frac{1}{2\sqrt{2}}$ gives

(4). Verified independently by direct numerical double integration and by symbolic integration in a computer algebra system, agreeing to floating-point precision throughout. □

Example. The $(1,1)$ cell ($x_0 = y_0 = 0$) has exact area $A_{\text{cell}}(0,0) = \frac{2\sqrt{2}}{3} \approx 0.9428$. The $(2,6)$ cell used as the worked example in [1] ($x_0 = 2, y_0 = 6$) has exact area $A_{\text{cell}}(2,6) \approx 0.7919$. For cells away from the coordinate axes the centroid approximation $A_{\text{cell}}(x_0, y_0) \approx T_c/(\sqrt{2}\,Y_c)$, evaluated at the cell's own center $(x_0 + \frac{1}{2}, y_0 + \frac{1}{2})$, is already accurate — 0.7894 against the exact 0.7919 for the $(2,6)$ cell — but it fails badly for the $(1,1)$ cell, giving $1/\sqrt{2} \approx 0.7071$ against the exact 0.9428, since that cell touches the coordinate axes where the density $(2')$ is unbounded near $x = 0$ or $y = 0$ even though it stays finite and equal to $1/\sqrt{2}$ along the diagonal itself; the cell average is pulled well above the centroid value by that nearby singularity. Proposition 4 is exact in both regimes, since it integrates $(2')$ directly rather than approximating it as constant.

Writing $K = 2j + 1$, so $K + 1 = 2j + 2$, (2) becomes exactly

$$A_{\text{band}}(j) = \frac{\pi}{\sqrt{2}}(2j + 2) = \sqrt{2}\,\pi\,(j + 1) \;\sim\; \sqrt{2}\,\pi\,j \quad (j \to \infty).$$

Comparing against the large-j limit of (1), $A(j) \approx 8\pi\gamma\ell_P^2 j$, and setting $\ell_P \equiv 1$ (one lattice unit identified with one Planck length):

$$\sqrt{2}\,\pi\,j = 8\pi\gamma\,j \quad\Longrightarrow\quad \gamma_{\text{eff}} = \frac{1}{4\sqrt{2}} \approx 0.1768. \tag{3}$$

K	$j = (K-1)/2$	$A_{\text{band}}(K)$, eq. (2)	implied $\gamma = A_{\text{band}}/(8\pi j)$
10	4.5	24.436	0.216
30	14.5	68.865	0.189
100	49.5	224.366	0.180
1000	499.5	2223.663	0.177

Table 1. Convergence of the implied γ to $\gamma_{\text{eff}} = 1/(4\sqrt{2})$.

3. What this does and does not show

Remark 5 (Genuine content)**.** *The functional form matches by construction: both (1) and (2) are, at leading order, linear in j, so any classically-computed area attached to a growing spin label will reproduce this shape. What is a nontrivial, checked fact is that the coefficient convergence in Table 1 is exact and monotone, following from Propositions 1–2 rather than from a numerically fitted discrete sum.*

Remark 6 (Which constants are forced, and which are stipulated)**.** *It should be stressed that γ_{eff} is not a free parameter tuned to fit: once the two identifications below are made, its value is forced by two facts that are themselves not free. The factor $\sqrt{2}$ in (2) is the cone's own $45°$ opening angle, fixed identically by $X^2+Y^2 = T^2$ before any physical input is introduced; the factor 8π in (1) is fixed by the normalization $\kappa = 8\pi G$ of the Einstein field equations underlying the Ashtekar–Barbero flux variables, not a convention internal to this note. The only genuine freedom in the entire calculation is the pair of stipulative identifications: (i) one lattice unit $\equiv$ one Planck length, and (ii) "the area of one edge" $\equiv$ this classical band. Given (i)–(ii), $\gamma_{\text{eff}} = 1/(4\sqrt{2})$ follows deterministically.*

Remark 7 (Running the identification the other way)**.** *Rather than fixing the lattice unit and solving for γ, one may fix γ at its physically determined value and solve (3) for the lattice unit L in Planck lengths: $L = \sqrt{4\sqrt{2}\,\gamma}\,\ell_P$. Using $\gamma \approx 0.2375$ [4] gives $L \approx 1.159\,\ell_P$; using $\gamma = \ln 2/(\pi\sqrt{3}) \approx 0.1274$ [5] gives $L \approx 0.849\,\ell_P$. Both land within a factor of order unity of ℓ_P itself without further* *adjustment, which is a mildly more suggestive fact than any particular value of* γ_{eff} *– though it remains a consequence of the two stipulative identifications above, not a derivation of the Planck scale from the multiplication table.*

Remark 8 (The unresolved scale mismatch, now quantitative)**.** *This calculation identifies a classical* band *of the cone – an entire two-row strip, containing on the order of* K *lattice cells – with a* single *LQG edge label* K*. It therefore does not resolve, and should not be read as resolving, the mismatch already noted informally: by Proposition 4, a single unit lattice cell carries only order-unity area (*$A_{\text{cell}}(0,0) \approx 0.94$*,* $A_{\text{cell}}(2,6) \approx 0.79$*, and in general* $A_{\text{cell}} = O(1)$ *for any fixed cell), while the band* $A_{\text{band}}(K) = \frac{\pi}{\sqrt{2}}(K+1)$ *of equation (2) grows linearly in* K *and contains on the order of* K *such cells: two genuinely different orders of coarse-graining, not merely an unquantified qualitative gap. So "one cell" and "one puncture" remain different objects at different levels of coarse-graining. The correspondence developed here is between an edge label and a* band*, not between a lattice cell and a puncture.*

4. Conclusion

The exact band-area formula (2), following from Propositions 1 and 2, matches the large-j growth rate of the loop quantum gravity area spectrum (1) up to an overall constant. Fixing that constant by an otherwise-unmotivated choice of units yields $\gamma_{\text{eff}} = 1/(4\sqrt{2})$, a number of the right order of magnitude but with no independent physical justification. We record this as a fully worked, self-contained numerical curiosity attached to the cone construction of [1, 2], useful chiefly as a semiclassical comparison would have to be set up rigorously, and explicitly not as a claim of any relation between the multiplication table and quantum gravity.

References

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